

MEDIATEK

MT8676 Hypervisor GPIO User Manual

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Version History

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1 GPIO

1.1 Overview

1.1.1 Brief Introduction

This chapter introduces the hardware features, software configuration and functions of GPIO controller, also contains the debugging methods for common problems.

1.1.2 Abbreviations

Table 1-1. Abbreviations

Abbreviation	Explanation
DATAIN	The value read by the pin is valid only when IES = 1.
DATAOUT	Set output High/Low in GPIO mode
DIR	Set Pin input/output in GPIO mode
DRV	General driving settings for the pin
EH	I2C-type pin-specific driving settings
IES	Input enable
MODE	Pinmux settings, just fill in the mode number
PD	When the pull-down is enabled for some pins, the PU must be set to 0.
PU	When the pull-up is enabled for some pins, the PD must be set to 0.
PUPD	Pull up/down settings for some pins with adjustable internal pull-up and pull-down resistances
R0/R1	Enable pull-up and pull-down resistors (available for some pins)
RSEL	Adjust pull-up and pull-down resistances for some pins (available for some pins)
SMT	Waveform filtering function

1.2 Architecture/Process Overview

1.2.1 GPIO Introduction

GPIO means General-Purpose Input/Output. In embedded systems, GPIO is an interface for digital signal interaction with external devices. It can read or control the status of external devices by setting it to input or output mode. MediaTek SoC provides PIN Controller hardware unit to implement:

- Pin function configuration. For example, whether the I/O pin is a normal GPIO or some special function pin (such as the CMD signal on the EMMC).
- Pin characteristic configuration. For example, the setting of pull-up/down resistance, the setting of drive-strength.

Provide GPIO Controller hardware unit to:

- Configure the direction of GPIO.
 - If it is output, you can configure it as high level or low level.
 - If it is input, you can get the level status on the GPIO pin.

1.2.2 Pinctrl Subsystem and GPIO Subsystem

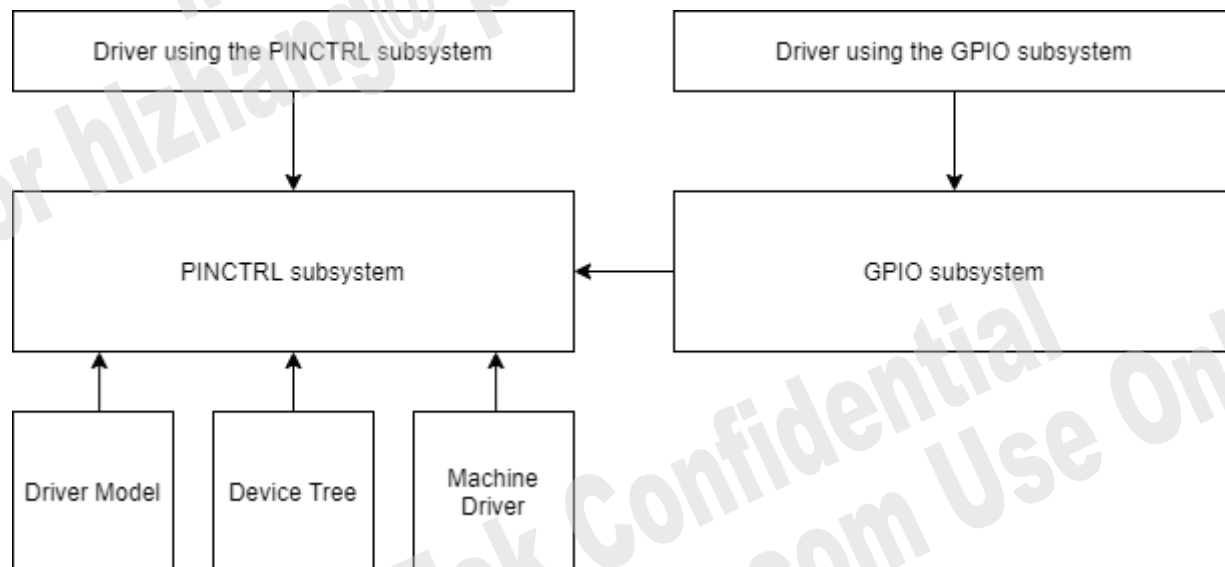


Figure 1-1. Pinctrl subsystem and GPIO subsystem

1.3 Configuration/Customization Guideline

1.3.1 Pinctrl Usage

The usage process of Pinctrl is as follows:

1. Configure the pin under the dts pio node, which needs to pay attention to the following points:
 - (1) The configured state name (mmc0_pins_default in the following example). The state name is the index keyword of the setting in dts, so be careful to identify it.
 - (2) Group the pins under the state (pin_cmd_dat {...} in the following example). Each group of pins must have the same pinconf setting. If there are different settings, multiple groups of settings can be written in one state.
 - (3) Configure pinmux (pinmux = <...> in the following example). Pinmux will describe the pin to be set and determine its function. The referenced definition comes from the file include/dt-bindings/pinctrl/mt6897-pinfunc.h.
 - (4) Configure pinconf (input-enable in the following example, etc.). After configuring pin config after pinmux, you can set the pull-up and pull-down states, input high/low, driving capability, input enable, etc.
2. Configuration under each module node in dts. The following are the references to pinctrl under each module node in dts:

```

/* dts Sample */
pinctrl-names = "default", "state_uhs";
pinctrl-0 = <&mmc0_pins_default>;
  
```

```
pinctrl-1 = <mmmc0_pins_uhs>;
```

3. Call in driver

```
/* C code Sample */
pinctrl *p = bri->pinctrl;
struct pinctrl_state = bri->pins_gpio

bri->pins_gpio = pinctrl_lookup_state(p, "gpio");
if (IS_ERR(bri->pins_gpio)) {
    dev_dbg(dev, "no gpio or recovery state found for GPIO recovery\n");
    bri->pins_gpio = NULL;
}
if (bri->pinctrl)
    pinctrl_select_state(bri->pinctrl, bri->pins_gpio);
```

1.3.2 GPIO Usage

The usage process of GPIO is as follows:

1. Configuration under the dts module node.

```
/* dts Sample */
&your_device_node {
    /* xx 代表 gpio number, e.g. xx = 155 */
    test-gpios=<&pio xx GPIO_ACTIVE_HIGH>;
};
```

2. Call in driver

```
/* C code Sample */
#include <linux/gpio.h>
#include <linux/consumer.h>
struct gpio_desc *gpio_spec;

gpio_spec = devm_gpiod_get(&pdev->dev, "test", GPIOD_OUT_HIGH);

/* Set direction to input mode */
gpiod_direction_input(gpio_spec);

/* get input value */
gpiod_get_value(gpio_spec);

/* Set direction to output mode and output value*/
gpiod_direction_output(gpio_spec, 1 or 0);

gpiod_set_value(gpio_spec, 1 or 0);

/* Free this GPIO if needed*/
devm_gpiod_put(&pdev->dev, gpio_spec);
```

1.4 Frequently Asked Questions/Troubleshooting

1.4.1 How to Check and Modify the PIN Status in Kernel

Use the mtk_gpio debug node to check and modify the PIN status.

```
#: cat /proc/mtk_gpio/soc.pinctrl
PIN: [MODE] [DIR] [DOUT] [DIN] [PULL_EN] [PULL_SEL] [IES] [SMT] [DRIVE] ( [R1] [R0] )
0: 0 0 0 0 1 0 1 0 0
1: 0 0 0 0 1 0 1 0 0
2: 0 1 1 1 1 0 1 0 0
3: 6 0 0 0 1 0 1 0 0
```

Description:

- MODE: Aux.Function selection, range: 0~7
- DIR: 0 for input mode; 1 for output mode (this is register value of MediaTek's DIR bit)
- DOUT/DIN: 0 for low; 1 for high
- DRIVE: Driving current selection, range: 0/1/2/3/4/5/6/7
- IES/SMT: 0 for disable; 1 for enable
- PULLEN/R1/R0: 0 for disable; 1/2/3 for enable
 - For pin with 2 pull resistors, R1 and R0 are shown.
 - For pin with 1 pull resistor, R1 and R0 are not shown.
- PULLSEL: 0 for selecting pull-down resistor; 1 for pull-up resistor

1.4.2 The Pin Level during Power-on Does Not Match the Default Reset Value

1. First check whether the dws file has configured this pin to a non-reset default state.
2. Then check whether there is any peripheral circuit impact.
3. Check whether preloader/lk has actively called the GPIO interface to set it.

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